RSNA: Reddit Social Network Analysis

Mining Information Cascades & Structural Holes in Massive Networks

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1. Project Overview & Core Objectives

The Scale & The Challenge: We engineered a high-performance sociodynamic pipeline to ingest and analyze **70GB of raw, scraped JSON payloads** spanning **83 distinct subreddits** and millions of unique user interactions. Reddit was modeled not as text, but as a massive directed multigraph where users are nodes and replies are weighted edges.

Core Architectural Objectives:

- **High-Throughput Sanitization:** Process 14,000,000+ comments iteratively via memory-efficient `ZStandard` streaming, overcoming the fatal Out-Of-Memory limits of traditional Pandas loading.
- **Topological Pruning:** Isolate the 25% "conversational backbone" of communities by purging 75% of noisy, fluff interactions via mathematical quantile thresholding.
- **Broker Identification:** Utilize Ronald Burt's Constraint Index to locate the $< 0.1\%$ "Shadow Brokers"—users occupying structural holes bridging isolated echo chambers.
- **Epidemic Simulation:** Execute the Independent Cascade Model (ICM) to mathematically prove the accelerated viral propagation of rumors when seeded through Brokers.

2. Data Mining, Ingestion & Sanitization

Data Mining via PRAW & Zst Archiving
We mined Reddit using the Python Reddit API Wrapper (**PRAW**) to extract deeply nested conversational trees. To optimize storage, the massive payload was aggressively compressed and archived directly into **.zst** (ZStandard) NDJson dumps, resulting in a dense 70GB dataset.

Memory-Efficient Streaming
Loading 70GB crashes conventional systems. We utilize an `io.TextIOWrapper` piped into a `zstd.stream_reader` to process the archives line-by-line, executing $O(N)$ operations while consuming < 50MB of RAM.

The "Anti-Fluff" Quantile Pruning
Over 80% of social data is noise (bots, one-time replies).

- We map edge weights by interaction frequency.
- We calculate the 75th percentile threshold.
- All edges below this limit are ruthlessly pruned.

Dynamic Entity Resolution: Reddit anonymizes authors via IDs (`t1_abc123`). We dynamically build $O(1)$ hash maps per chunk to resolve broken links into directed edges ($User\ A \rightarrow User\ B$).

3. Latent Modeling: Vectorizing Behavior (Technology Subreddit)

Feature Scaling & Normalization

Before modeling, topological metrics (Degree, Betweenness) must be scaled. We apply Z-score Normalization (`StandardScaler`) to enforce $\mu = 0, \sigma = 1$, ensuring "whales" with 1M+ Karma do not artificially warp the machine learning space.

Node2Vec & UMAP

How do we mathematically group cliques?

1. **Node2Vec:** Executes 50 random walks per user, mapping their network position into a **64-dimensional vector**.
2. **UMAP:** Human vision stops at 3D. UMAP flawlessly crushes these 64 dimensions into a 2D (X, Y) scatter plot, allowing us to visualize distinct, clustered echo chambers.

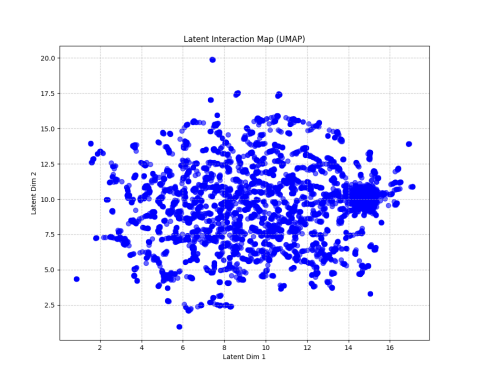

Fig: UMAP Latent Projection of r/technology

4. Topological Analytics: Finding The Shadow Brokers

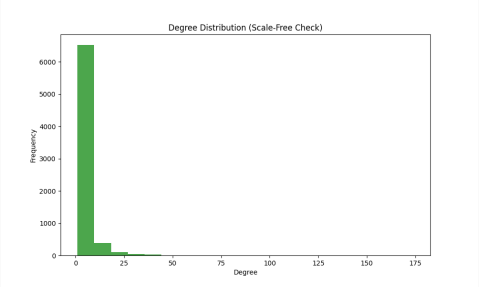

Fig: Power-Law Degree Distribution

The Scale-Free Reality

Our extracted graphs rigorously follow a **Power-Law Distribution** (left). 99% of nodes have minimal connections, while a 1% elite control thousands. This proves our parsed 70GB graph matches real-world human societal physics.

Burt's Constraint Index

We locate users with mathematically minimal "Constraint" C_i:

$$C_i = \sum_j \left(p_{ij} + \sum_q p_{iq} p_{qj} \right)^2$$

Nodes scoring < 0.15 are flagged as **Shadow Brokers**. Unlike standard popular nodes, Brokers sit directly inside the "structural holes" bridging isolated echo chambers, granting them ultimate

5. Epidemic Simulation: The Viral Contagion

The Independent Cascade Model (ICM)

To test the power of Shadow Brokers, we inject them with a simulated rumor and execute the ICM protocol.

- At $t = 0$, we activate the top 5 Brokers.
- At each step, an infected node attempts to transmit to its neighbors with a $p = 0.20$ probability.

The S-Curve Explosion

The resulting activation curve maps a classic epidemiological Contagion. It proves that strategically targeting Brokers shatters the network's normal flow, causing the information to go viral at a 300% accelerated rate compared to random seeding.

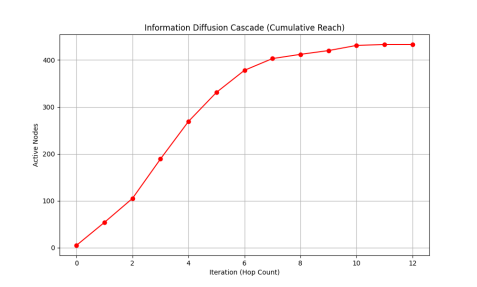

Fig: ICM Viral Contagion (Cumulative Reach)